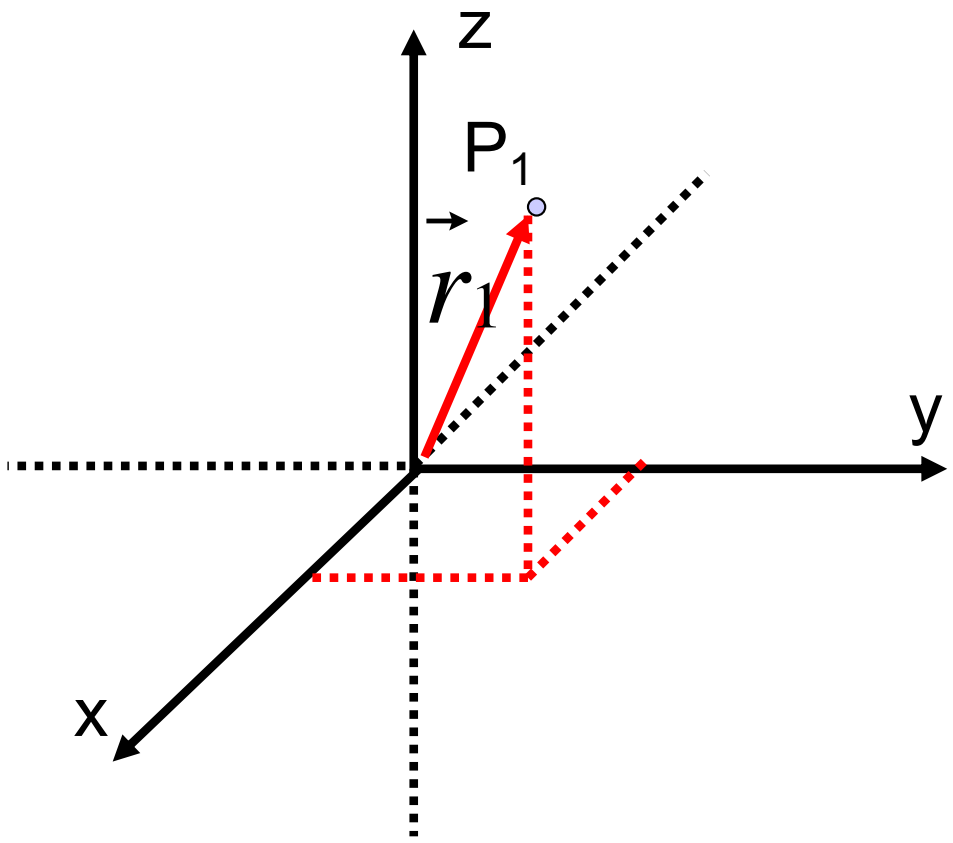
(Review at home)

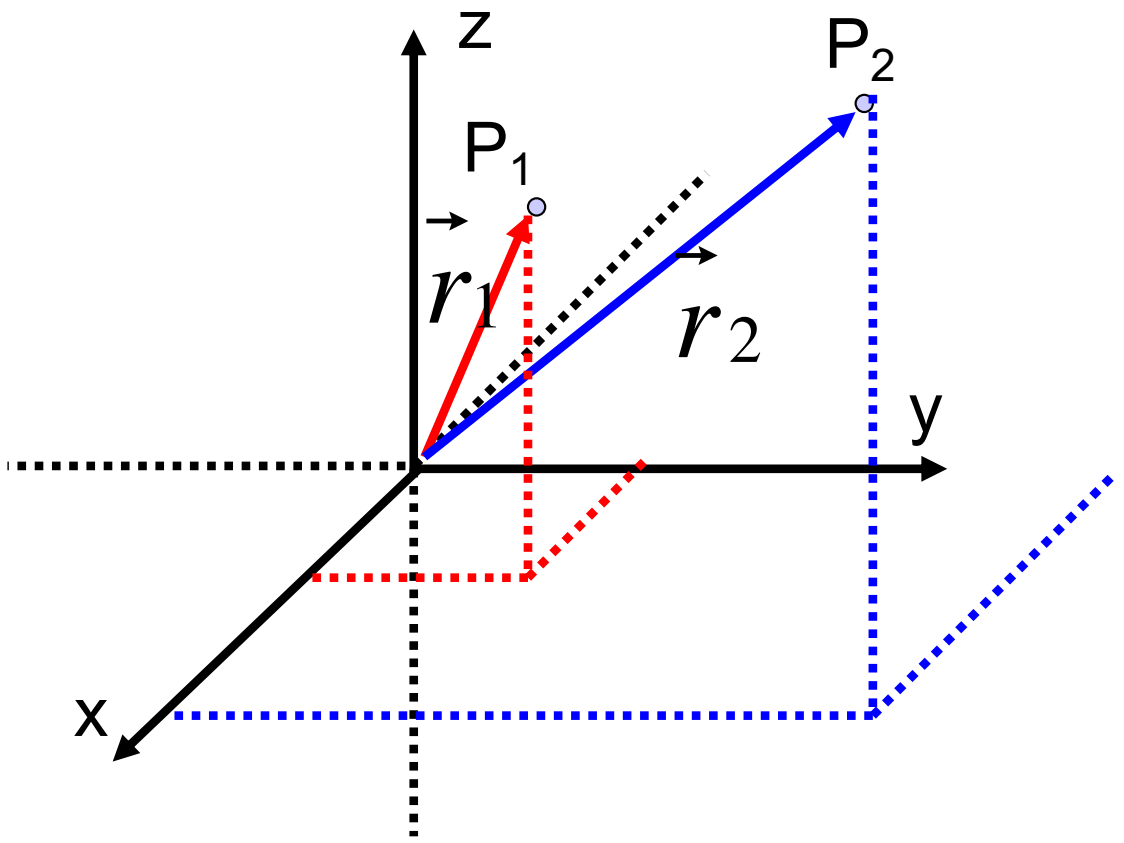
Vector From Origin to a Point



P_1: (x_1, y_1, z_1)

(Review at home)

Vector Between Two Points

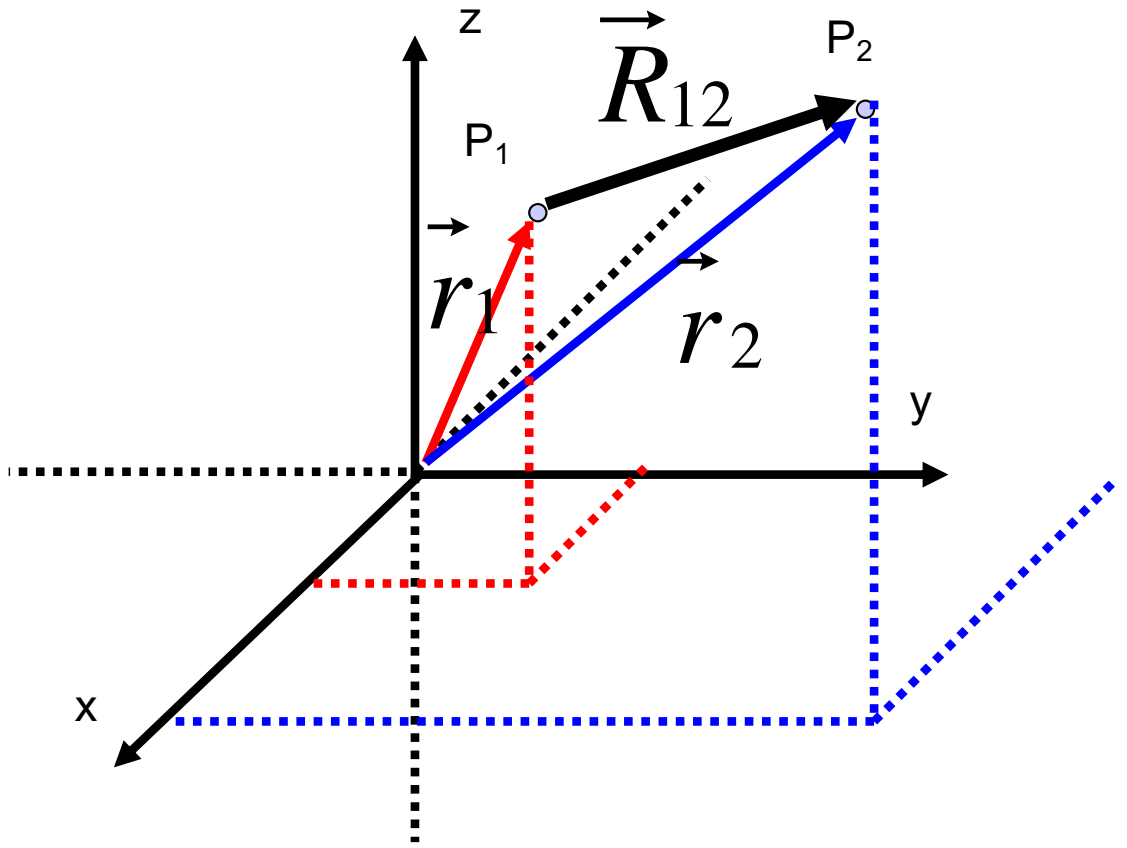


P_1: (x_1, y_1, z_1)

P_2: (x_2, y_2, z_2)

(Review at home)

Vector Between Two Points



$$\vec{R}_{12} = (FinalPosition) - (InitialPosition)$$

$$\vec{R}_{12} = (x_2 - x_1)\hat{a}_x + (y_2 - y_1)\hat{a}_y + (z_2 - z_1)\hat{a}_z$$

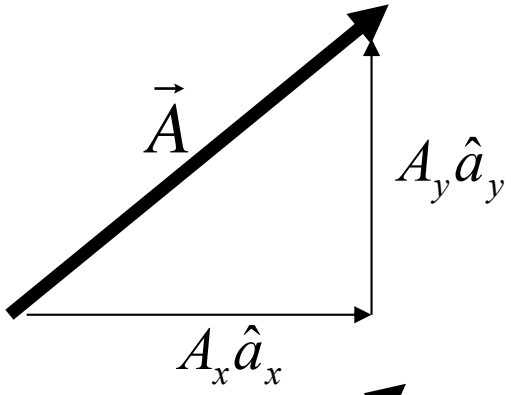
(Review at home)
Vector Math

$$\vec{A} = A_x\hat{a}_x + A_y\hat{a}_y + A_z\hat{a}_z$$

$$\vec{B} = B_x\hat{a}_x + B_y\hat{a}_y + B_z\hat{a}_z$$

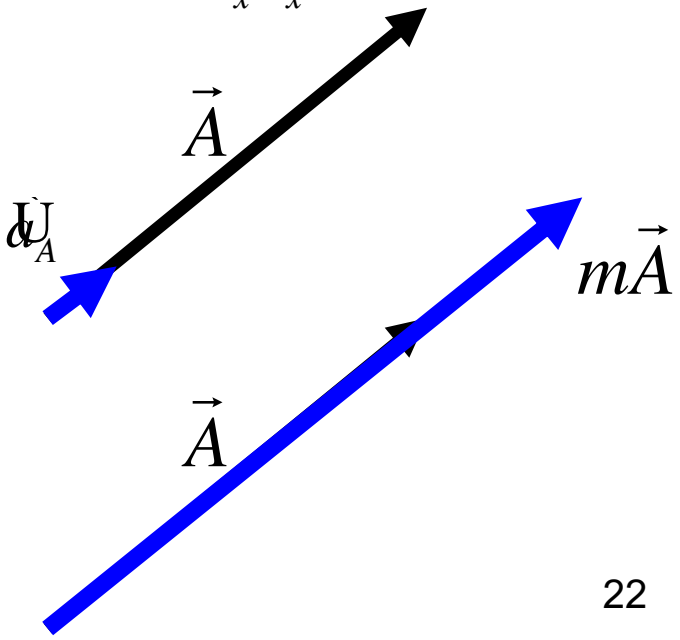
Magnitude of **A**

$$\left|\vec{A}\right| = \sqrt{A_x^2 + A_y^2 + A_z^2}$$



Unit vector in direction of **A**

$$\hat{a}_A = \frac{\vec{A}}{\left|\vec{A}\right|} = \frac{A_x\hat{a}_x + A_y\hat{a}_y + A_z\hat{a}_z}{\sqrt{A_x^2 + A_y^2 + A_z^2}}$$

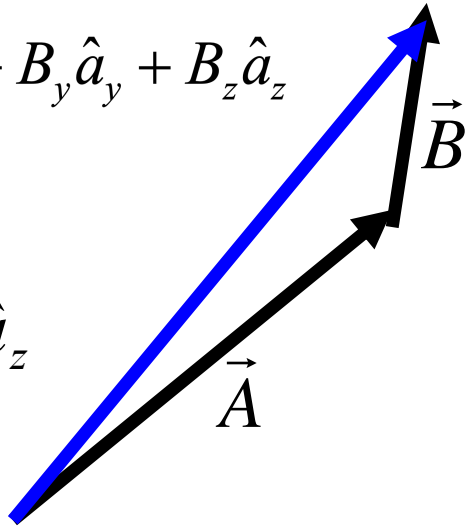


Multiplication of **A** by a scalar

$$m\vec{A} = mA_x\hat{a}_x + mA_y\hat{a}_y + mA_z\hat{a}_z$$

(Review at home)
Vector Math

$$\vec{A} = A_x\hat{a}_x + A_y\hat{a}_y + A_z\hat{a}_z \qquad \vec{B} = B_x\hat{a}_x + B_y\hat{a}_y + B_z\hat{a}_z$$

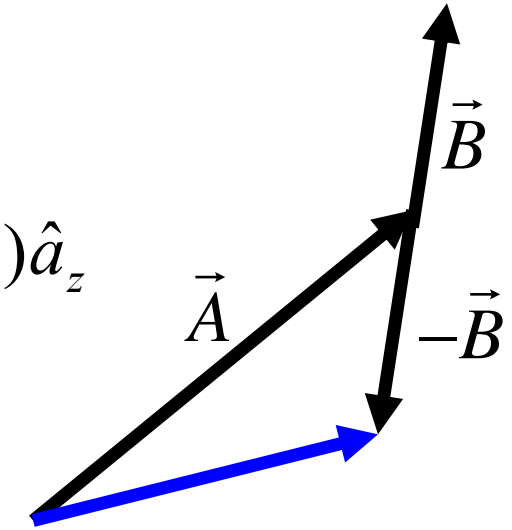


Addition of **A** and **B**

$$\vec{A} + \vec{B} = (A_x + B_x)\hat{a}_x + (A_y + B_y)\hat{a}_y + (A_z + B_z)\hat{a}_z$$

Subtraction of **A** and **B**

$$\vec{A} - \vec{B} = (A_x - B_x)\hat{a}_x + (A_y - B_y)\hat{a}_y + (A_z - B_z)\hat{a}_z$$

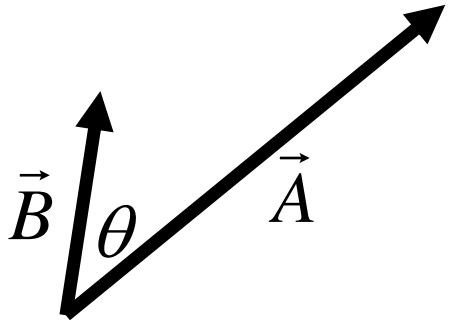


(Review at home)
Dot Product

$$\vec{A} = A_x\hat{a}_x + A_y\hat{a}_y + A_z\hat{a}_z \qquad \vec{B} = B_x\hat{a}_x + B_y\hat{a}_y + B_z\hat{a}_z$$

Dot Product of **A** and **B**

$$\vec{A} \bullet \vec{B} = \left|\vec{A}\right|\left|\vec{B}\right|\cos\theta$$



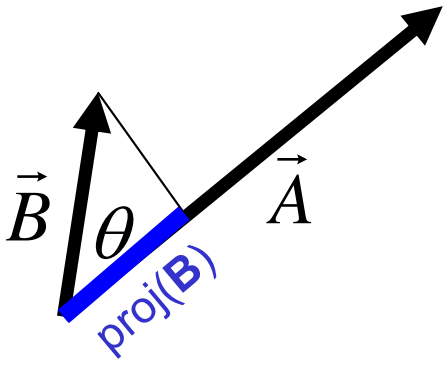
Resulting quantity is a SCALAR

Physical Meaning

(Magnitude of **A**)*(Projection of **B** onto **A**)

or (Magnitude of **B**)*(Projection of **A** onto **B**)

$$\vec{A} \bullet \vec{B} = \vec{B} \bullet \vec{A}$$



(Review at home)

Dot Product

$$\vec{A} = A_x \hat{a}_x + A_y \hat{a}_y + A_z \hat{a}_z \qquad \vec{B} = B_x \hat{a}_x + B_y \hat{a}_y + B_z \hat{a}_z$$

Dot Product of **A** and **B**

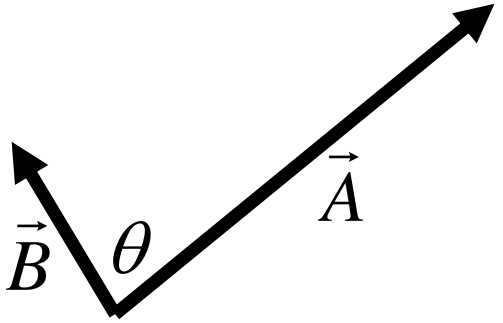
$$\vec{A} \bullet \vec{B} = \left|\vec{A}\right|\left|\vec{B}\right| \cos\theta$$

$\vec{A} \perp \vec{B}$

$$\cos\theta = 0$$

$$\vec{A} \bullet \vec{B} = 0$$

$$\vec{A} \perp \vec{B} \Leftrightarrow \vec{A} \bullet \vec{B} = 0$$

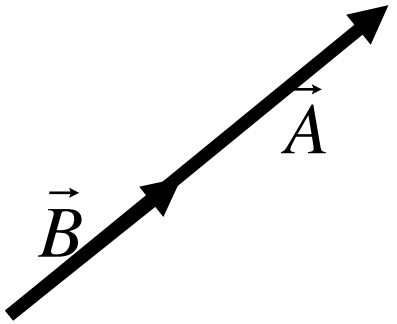


$\vec{A} \| \vec{B}$

$$\cos\theta = 1$$

$$\vec{A} \bullet \vec{B} = \left|\vec{A}\right|\left|\vec{B}\right|$$

$$\vec{A} \parallel \vec{B} \Leftrightarrow \vec{A} \bullet \vec{B} = \left|\vec{A}\right|\left|\vec{B}\right|$$



(Review at home)

Dot Product

$$\vec{A} = A_x\hat{a}_x + A_y\hat{a}_y + A_z\hat{a}_z \qquad \vec{B} = B_x\hat{a}_x + B_y\hat{a}_y + B_z\hat{a}_z$$

$$\vec{A} \perp \vec{B} \Leftrightarrow \vec{A} \bullet \vec{B} = 0$$

$$\vec{A} \parallel \vec{B} \Leftrightarrow \vec{A} \bullet \vec{B} = \left|\vec{A}\right|\left|\vec{B}\right|$$

Dot Product of **Unit Vectors**

$$\hat{a}_x \bullet \hat{a}_x = 1 \quad \hat{a}_y \bullet \hat{a}_x = 0 \quad \hat{a}_z \bullet \hat{a}_x = 0$$
$$\hat{a}_x \bullet \hat{a}_y = 0 \quad \hat{a}_y \bullet \hat{a}_y = 1 \quad \hat{a}_z \bullet \hat{a}_y = 0$$
$$\hat{a}_x \bullet \hat{a}_z = 0 \quad \hat{a}_y \bullet \hat{a}_z = 0 \quad \hat{a}_z \bullet \hat{a}_z = 1$$

$$\vec{A} \bullet \vec{B} = \left(A_x\hat{a}_x + A_y\hat{a}_y + A_z\hat{a}_z\right) \bullet \left(B_x\hat{a}_x + B_y\hat{a}_y + B_z\hat{a}_z\right)$$

$$\vec{A} \bullet \vec{B} = A_xB_x + A_yB_y + A_zB_z$$

Scalar